(11.)

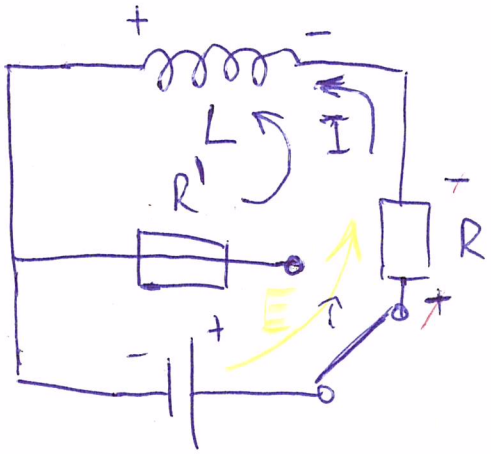
RL ~~and~~ RC and RLC circuits

sw on first !!

$-\int_1^2 \underline{E}_i \, d\underline{l} < 0$



I. Switching off an RL circuit



$I_0 = \frac{U_0}{R}$ (The inductance is ∅ at DC)

switching off (to R') at $t = 0$

$t \geq 0$ Kirchhoff 2: $U_R + U_L = 0$

$$-I(R+R') + L\frac{dI}{dt} = 0$$

(−) potential drops when going ∥ with I

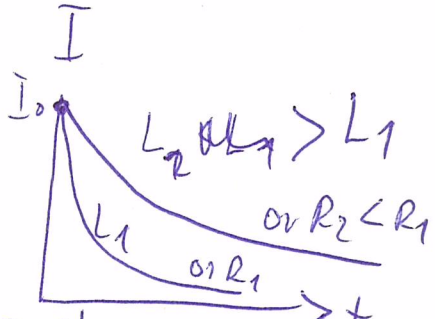
(+) induced voltage acts against the reduction of the current, so + $\left(\frac{dI}{dt} < 0\right)$

$$\frac{dI}{dt} = -\frac{R+R'}{L} I$$

$$\Downarrow$$

$$I(t) = I_0 \cdot e^{-\frac{R+R'}{L} \cdot t}$$





$$U_{ind} = L\frac{dI}{dt} = \frac{U_0}{R} \cdot \frac{R+R'}{L} \cdot e^{-\frac{R+R'}{L} t} = U_0 \cdot \frac{R+R'}{R} \cdot e^{-}$$

Can be substantially larger than U_0 !!!

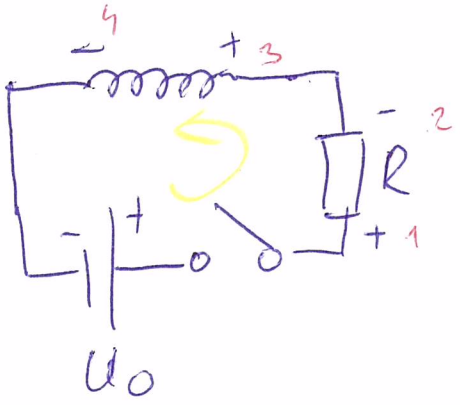
(if $R' \gg R$ → very fast voltage drop → high ind. voltag

air, resistor

↳ electric discharge !

Sparks can be produced @ switch off !!

II Switching on an RL circuit



$I(t<0) = 0$, $t=0$ switch on

$t \geq 0$ $U_R + U_L + U_0 = 0$ Kirch 2.

$$-IR - L\frac{dI}{dt} + U_0 = 0$$

$$\frac{dI}{dt} = \frac{U_0 - IR}{L}$$

$$\frac{dI}{U_0 - IR} = \frac{dt}{L}$$

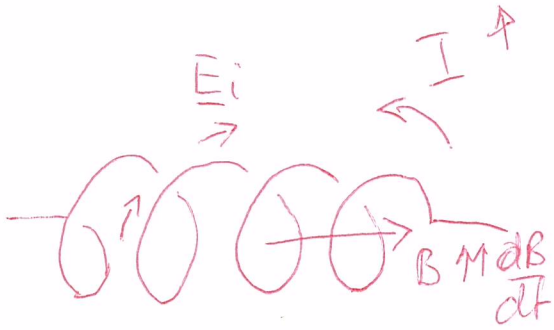
$$\int_0^{I(t)} \frac{dI}{U_0 - IR} = \int_0^t \frac{dl}{l}$$

$$-\frac{1}{R}\ln\left(\frac{U_0 - IR}{U_0}\right) = \frac{t}{L}$$

$$\frac{U_0 - IR}{U_0} = e^{-\frac{R}{L}t}$$

$$U_0 - IR = U_0 e^{-\frac{R}{L}t}$$

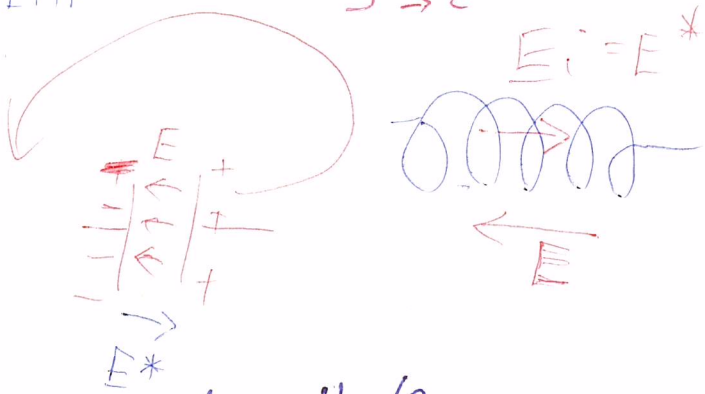
$$I = \frac{U_0\left(1 - e^{R/L \cdot t}\right)}{R}$$

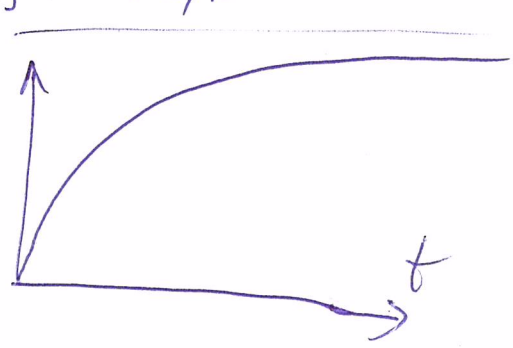


E_0 ,

$\mathcal{E} = U_L = -\int E_i\, dL > 0$

EMF $3 \to 2$

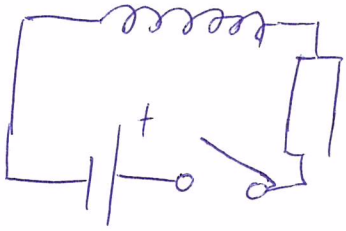




Energy of the magnetic field

Experience: when switching on and off a circuit containing inductivity L, the change of current (and magnetic field ~ current) can not be instantaneous. The electro-magnetic induction is maintaining the current of the circuit. It needs energy for this!

Let's investigate again the switching on!



$$U_0 - IR - L\frac{dI}{dt} = 0 \qquad / \cdot I\,dt$$

$$U_0 \cdot I\,dt - I^2 R\,dt - LI\frac{dI}{dt}\cdot dt = 0$$

$U_0 \cdot I\,dt$: Work made by the battery

$I^2 R\,dt$: Joule-heat

$LI\frac{dI}{dt}\cdot dt$: ?? change of energy of magn. field.

Conservation of energy:

$$dE_{magn} = LI\frac{dI}{dt}\cdot dt = LI\,dI$$

$$E_{magn} = \int_0^I LI'\,dI' = \frac{1}{2}LI^2$$

$\int_0^I LI\frac{dI}{dt}\,dt$

Energy stored in an coil (inductor) of L self inductance.

Let us express it with magnetic field!

For a solenoid $B = \frac{\mu N I}{l} \rightarrow I = \frac{Bl}{\mu N}$

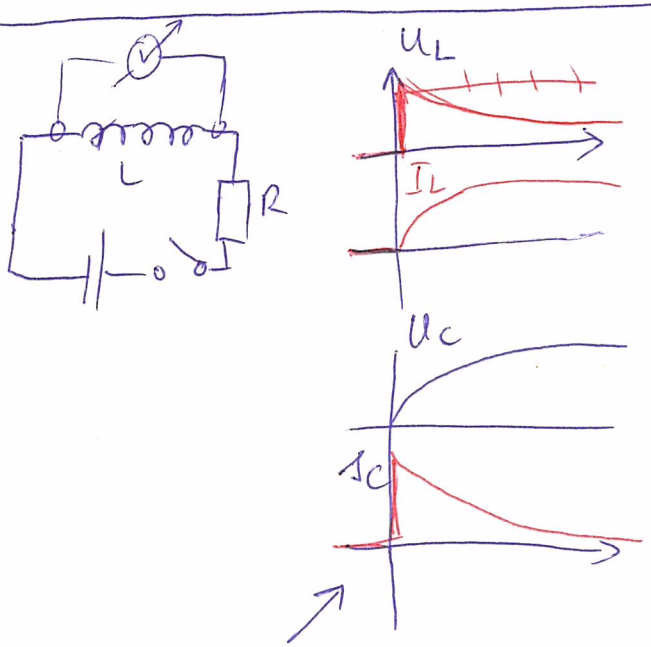
Self inductance of a solenoid: $L = \frac{\mu N^2 A}{l}$

$$E_{magn} = \frac{1}{2} L I^2 = \frac{1}{2} \frac{\mu N^2 A}{l} \cdot \frac{B^2 l^2}{\mu^2 N^2} = \frac{1}{2} \frac{B^2}{\mu} \cdot \underbrace{l \cdot A}_{V_{coil}}$$

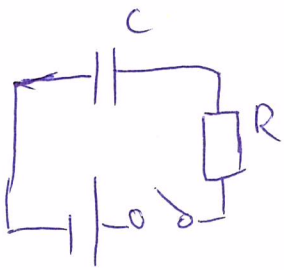
Energy density of magn. field:

$$w_{magn} = \frac{1}{2} \frac{B}{\mu} = \frac{1}{2} \underline{H} \cdot \underline{B} \qquad (\underline{B} = \mu \underline{H} \text{ for linear})$$

$$w_{el} = \frac{1}{2} \underline{D} \underline{E}$$



Because energy is stored in the el. field $\sim U$



$$U_0 - IR - \frac{Q}{C} = 0 \quad / \frac{d}{dt}$$

$$-\frac{dI}{dt} R - \frac{dQ}{dt} \frac{1}{C} = 0$$

$\frac{dQ}{dt} = -I$

$$\frac{dI}{dt} = -\frac{I}{RC}$$

$$\Downarrow$$

$$I(t) = I_0 e^{-\frac{t}{RC}}$$

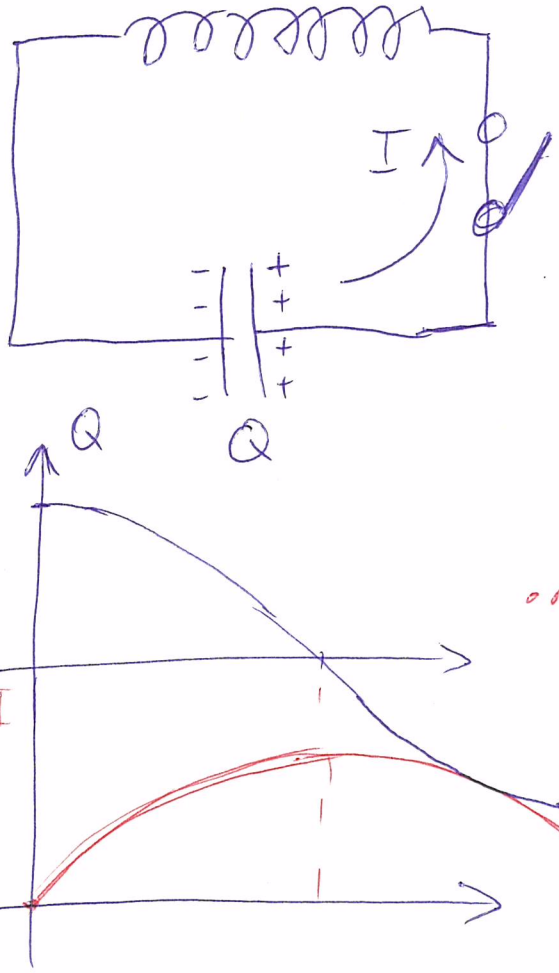
$I_0 = I(t=0)$ from initial cond: $Q(t=0) = 0$

$$U_0 - IR = 0$$

$$\downarrow$$

$$I_0 = \frac{U_0}{R}$$

LC circuit



Let's charge the capacitor and close the circuit!

- Capacitor start to discharge
- Current in the circuit
- growing current → EMI
- against the change!
- current grows until capacitor discharged
- Capacitor would stop, but L is against the change!
- EMI changes the C at reverse polarity

Quantitatively:

Kirchhoff II.

$$\frac{Q}{C} + -L\frac{dI}{dt} = 0$$

$$\frac{Q}{C} + L\frac{d^2Q}{dt^2} = 0 \rightarrow \frac{d^2Q}{dt^2} = -\frac{1}{LC}\cdot Q$$

Kirchhoff I

$$I(t) = -\frac{dQ}{dt}$$

solution: $Q(t) = Q_0 \cdot \sin(\omega_0 t + \varphi_0)$

substitute: $\frac{d^2Q}{dt^2} = Q_0 \cdot -\omega_0^2 \sin(\omega_0 t + \varphi_0) = -\frac{1}{LC} Q_0 \sin(\quad)$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{LC}} \qquad \left(f_0 = \frac{1}{2\pi}\omega_0 \right)$$ eigen (own) frequency / natural

Simple harmonic oscillation

Energy?

$$I(t) = \frac{-dQ}{dt} = -Q_0 \omega_0 \cos(\omega_0 t + \varphi_0)$$

$$E_{tot} = E_{el} + E_{magn} = \frac{1}{2}\frac{Q^2}{C} + \frac{1}{2}LI^2 =$$

$$= \frac{1}{2}\frac{Q_0^2}{C} \cdot \sin^2(\omega_0 t + \varphi_0) + \frac{1}{2} Q_0^2 \cdot \omega_0^2 \cos^2(\omega_0 t + \varphi_0)$$

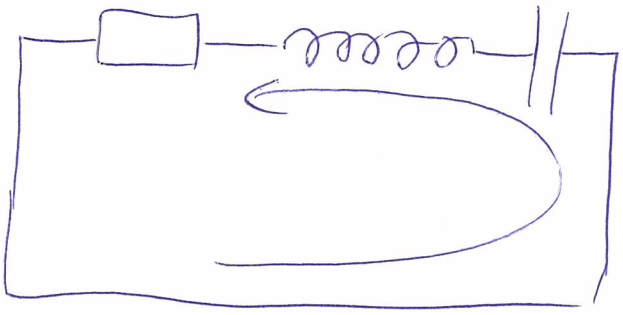
$$\omega_0^2 = \frac{1}{LC}$$

$$= \frac{1}{2}\frac{Q_0^2}{C}\left(\sin^2 + \cos^2\right) = \frac{1}{2}\frac{Q_0^2}{C} = \frac{1}{2} L I_0^2 = \text{const}!$$

Total energy const, energy oscillates back-forth between C and L.

RLC circuit (serial)

$\left(I = \frac{-dQ}{dt}\right)$



$$\frac{Q}{C} - L\frac{dI}{dt} - IR = 0$$

$I = \frac{-dQ}{dt}$ ↓ $\qquad$ $\frac{d}{dt}$ ↓

$$L\frac{d^2Q}{dt^2} + \frac{dQ}{dt}R + \frac{Q}{C} = 0 \qquad L\frac{dI}{dt} + IR + \frac{I}{C} = 0 \;/\; \frac{d}{dt}$$

$$L\frac{d^2I}{dt^2} + \frac{dI}{dt}R + \frac{I}{C} = 0$$

same equation for Q and I

resembles to the damped oscillation in mech.!!

$$\frac{d^2I}{dt^2} + \frac{R}{L}\frac{dI}{dt} + \frac{1}{LC}I = 0$$

$\frac{R}{L} \rightsquigarrow 2\beta$ $\qquad$ $\frac{1}{LC} \rightsquigarrow \omega_0^2$

$$\frac{d^2I}{dt^2} + 2\beta\frac{dI}{dt} + \omega_0^2 I = 0 \quad \Big| \quad \text{Hint: } I_0 \cdot e^{\alpha t}$$

↓

$$\left(\alpha^2 + 2\beta\alpha + \omega_0^2\right) e^{\alpha t} \cdot I_0 = 0$$

↓ $\quad \emptyset$

$$\alpha_{1,2} = \frac{-2\beta \pm \sqrt{4\beta^2 - 4\omega_0^2}}{2} = -\beta \pm \sqrt{\beta^2 - \omega_0^2}$$

strong damping: $\beta > \omega_0 \rightarrow$ exponential decay!

weak damping: $\beta < \omega_0$

$\sqrt{\beta^2 - \omega^2} < 0$ sq root of a neg. number? imaginary

$$\alpha_{12} = -\beta \pm i\sqrt{\omega_0^2 - \beta^2} = -\beta \pm i\omega_1$$

$\omega_1 < \omega_0$

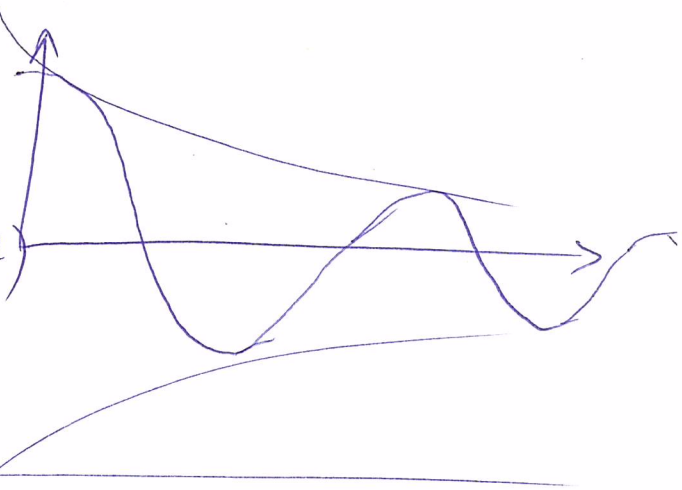
Substitute into hint:

$$I = I_0 \cdot e^{\alpha t} = I_0 e^{-\beta t} \cdot e^{i\omega_1 t}$$

physical solution is real: let's take the real part!

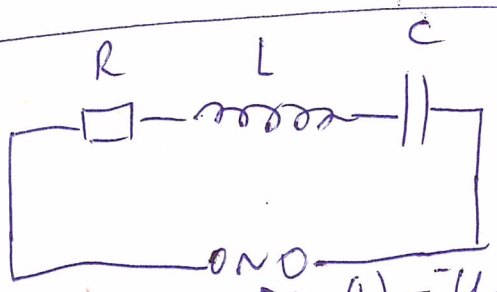
$$\mathrm{Re}(e^{i\omega_1 t}) = \cos(\omega_1 t)$$

$$I = I_0 e^{-\beta t} \cdot \cos(\omega_1 t)$$



Energy: dissipation (Joule heat) on resistor!

Forced oscillation of an RLC



$U_E(t) = U_0 \cdot \cos(\omega t)$ $\left(= \mathrm{Re}\ U_0 \cdot e^{i\omega t}\right)$

$U_R + U_L + U_C + U_E = 0$

$-IR - L\frac{dI}{dt} + \frac{Q}{C} + U_E = 0$ $\qquad -\frac{dQ}{dt} = I$

$\frac{dQ}{dt}R + L\frac{d^2Q}{dt^2} + \frac{Q}{C} + U_E = 0$

$\frac{dQ}{dt}R + L\frac{d^2Q}{dt^2} + \frac{Q}{C} = U_0 \cos\omega t$

$\frac{d^2Q}{dt^2} + \frac{R}{L}\frac{dQ}{dt} + \frac{Q}{LC} = \frac{U_0}{L}\cos\omega t$ $\qquad 2\beta$, ω_0^2

Solution
A. Gen. sol of hom. eq. (damped osc. disappears
B. long term oscillation with the ω of the ext. force

$\Leftarrow$ Hint: $Q(t) = \mathrm{Re}\left(Q_0(\omega)\, e^{i\omega t}\right)$

$-\omega^2 Q_0 e^{i\omega t} - 2\beta\omega i\, Q e^{i\omega t} + \omega_0^2 Q_0 e^{i\omega t} = \frac{U_0}{L} \cdot e^{i\omega t} \Rightarrow Q_0(\omega) = \frac{U_0/L}{-\omega^2 + 2i\beta\omega + \omega_0^2}$

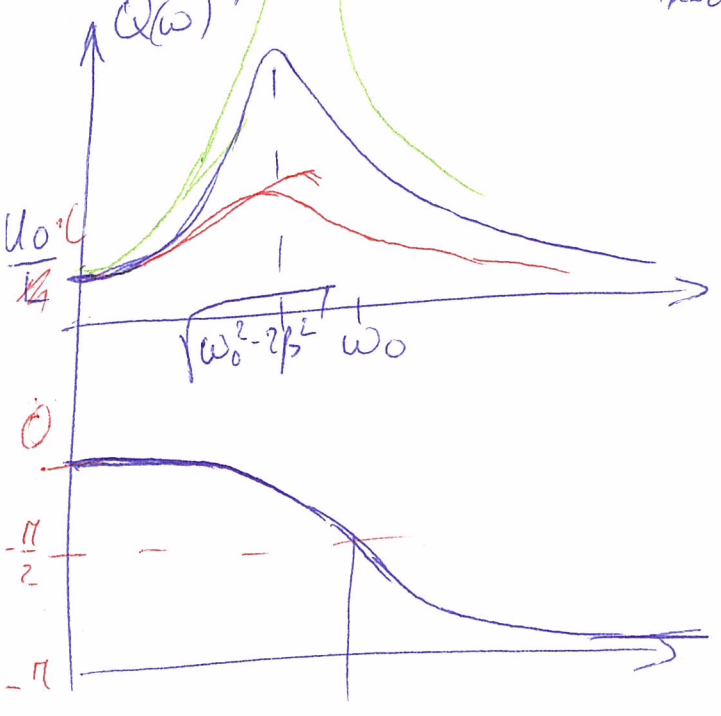
$|Q_0(\omega)| = \frac{U_0/L}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}}$

$\tan\varphi = \frac{2\beta\omega}{\omega_0^2 - \omega^2}$

$I = \frac{-dQ}{dt} = \underbrace{-i\omega Q(\omega)}_{\tilde{I}_0(\omega)} e^{i\omega t}$ $\qquad |I_0\omega| = |Q_0(\omega)| \cdot \omega$

$\varphi_I = \varphi_Q + \frac{\pi}{2}$

Voltage resonance (|| Amplitude resonance)



Current resonance (|| speed)

